The Illustrated Student's Guide to Self-Sorting Cells, "Platonic Space," and Why a 6-Line Algorithm Sparked a Huge Debate

How 100 Kids in Gym Class Explain Morphogenesis, Causal Emergence, and Mathematical "Free Lunches"

Joel Dietz
Infinite Morphospace Research Project · `https://github.com/fractastical/morpholib`

October 22, 2026

1 The Gym-Class Puzzle: What If There Is No Gym Teacher?

Imagine 100 kids standing in a line in gym class, each wearing a jersey with a random number from **1** to **100**. Normally, the gym teacher (a **central controller**) walks down the line and tells specific kids to swap places until everyone is lined up in order from 1 to 100.

In 2025, three researchers — **Taining Zhang, Adam Goldstein, and Michael Levin** (*Adaptive Behavior* 33(1):25–54) — asked a great question: **What if there is no gym teacher?** What if each kid is an independent "cell" who can only look at their neighbors and decide on their own whether to swap?

They gave the kids three different "personalities" (which they called **algotypes**, after classical computer-science sorting algorithms):

1. **Bubble kids (`B` — Blue):** Look at the kid immediately to your left or right. If your number is smaller than your left neighbor's (or bigger than your right neighbor's), swap places!
2. **Insertion kids (`I` — Green):** Only step left if everyone to your left is already in order.
3. **Selection kids (`S` — Orange):** Keep a mental finger pointing at slot #1, then #2, then #3... If your number is smaller than whoever is standing at that target slot, teleport straight there and swap with them!

When they ran computer experiments on this line of kids, they saw **three surprising behaviors**:

- **1. Sorting Around "Frozen" Kids:** If 2 or 3 kids refuse to move ("frozen defects"), the rest of the line still manages to sort itself!
- **2. "Delayed Gratification":** Sometimes, the line temporarily gets *more* messy (according to their messiness ruler) before getting much cleaner a few steps later.
- **3. "Secret Clique Clustering":** When you mix 50 **Bubble** kids and 50 **Selection** kids in the same line, the Bubble kids group together with other Bubble kids, and Selection kids group together with other Selection kids — **even though their jerseys don't show who is Bubble or Selection!**

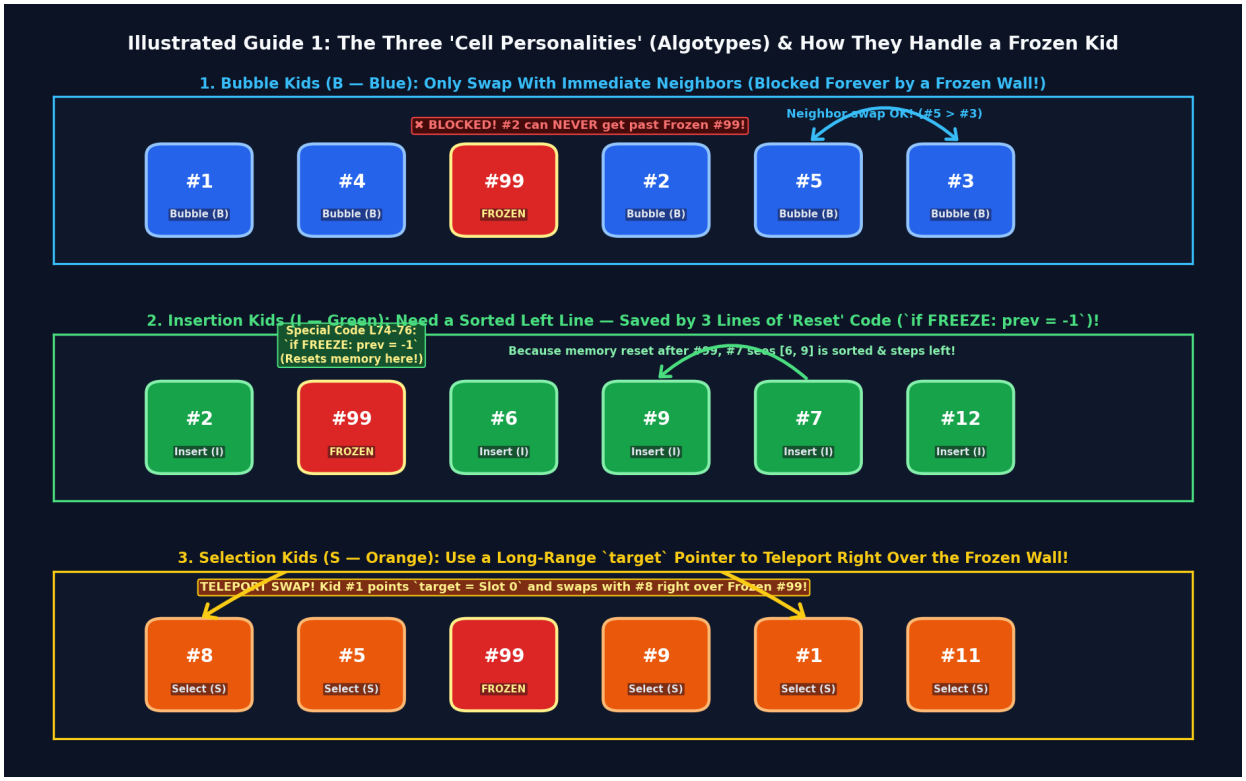


Figure 1: **The Three "Cell Personalities" (Algotypes) and How They Handle a Frozen Kid. Top (Bubble `B`):** Only swaps with immediate neighbors, so kid #2 is blocked forever behind Frozen #99. **Middle (Insertion `I`):** Normally halts if anyone to the left is out of order, but lines 74–76 of `InsertionSortCell.py` contain a 3-line reset rule (`if FREEZE: prev = -1`) that tells Insertion kids to start their check fresh right after a frozen kid. **Bottom (Selection `S`):** Uses a long-range `target` pointer to teleport right over Frozen #99.

2 The Big Debate: "Platonic Ghost Minds" vs. "Nothing to See Here!"

After that paper came out, a big debate broke out between two camps:

- **Camp 1 — Michael Levin's "Platonic Space" Claim (*Ingressing Minds*, 2025/2026):** Levin argued that because nobody wrote code telling the numbers how to handle broken cells, how to "temporarily get messier to go around a wall," or how to "find friends with the same personality," these behaviors must be **unprogrammed cognitive "side-quests"** beaming in from a non-physical **Platonic Space of Patterns** (the same mathematical realm where π and prime numbers live).

- **Camp 2 — The Computer-Science Blog Critics:** Several computer scientists wrote blog posts saying: *"Come on! That clustering is just a 1960s textbook trick called **Bubble-sort stability** (when two kids wear the exact same number, Bubble sort doesn't swap them). There's nothing new here at all!"*

To find out who was right, we read every single line of their original Python code, translated it into three other computer languages (**C++20**, **JavaScript**, and the **Lean 4** math theorem prover), and ran 3 "what-if" experiments (ablations) plus Erik Hoel's brand-new 2026 **Causal Emergence 2.0** math test. We discovered that **both camps got something right and something wrong!**

3 Discovery 1: Did They Really Add "Zero Code" for Frozen Kids?

In *Ingressing Minds*, Levin wrote that they *"did not add steps to the algorithm... to do specific things if an operation failed."*

- For **Bubble** kids, that is 100% true! If a Bubble kid tries to swap with a frozen kid, the frozen kid simply doesn't move.
- **Insertion** kids, however, have a strict rule: *"Never step left unless everyone to your left is already in order."* If a frozen kid wearing #99 is stuck near the front of the line (Figure 1, middle row), then the line to the left is *never* in order — so every single Insertion kid behind #99 would freeze forever!
- When we looked at lines 74–76 of their file `InsertionSortCell.py`, we found **3 lines of special code** that explicitly check `if status == FREEZE: prev = -1` (telling Insertion kids: *"If you see a frozen kid, pretend the line starts over right after them!"*). When we deleted those 3 lines of code, the Insertion kids froze up and their final error jumped **4.5**× higher (4.9 → 21.95)!

4 Discovery 2: Why "Delayed Gratification" Was an Optical Illusion of the Ruler!

Levin compared the sorting line to a person walking *backwards* around a wall to reach a goal (unlike two magnets stuck on opposite sides of a wooden board), because the paper's messiness score (**Neighbor Steps** $E(v)$) goes *up* (+1) 47 times before going down.

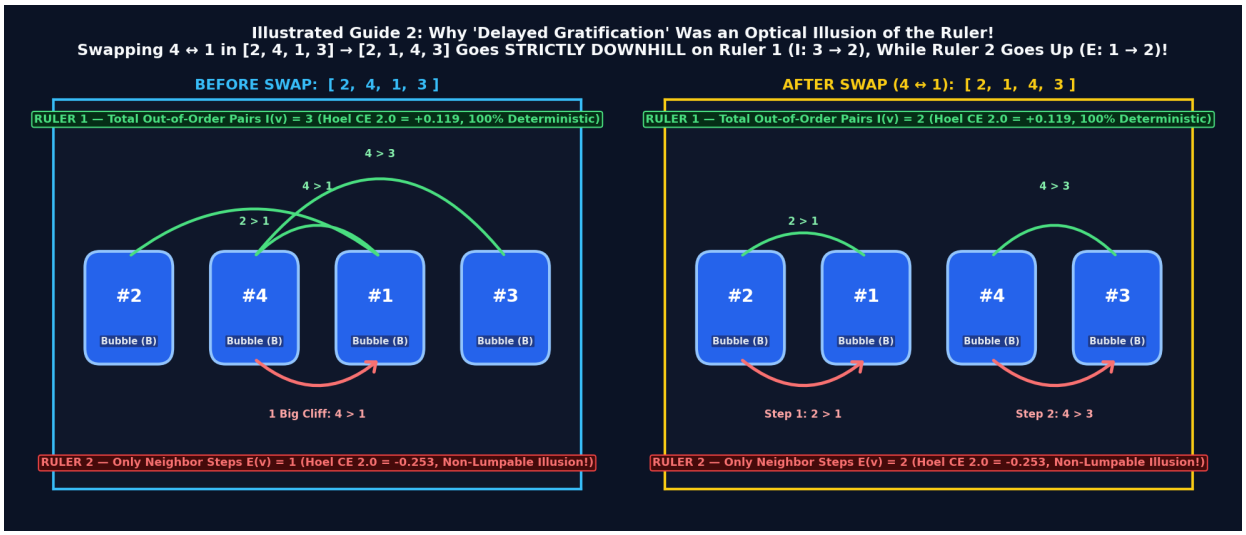


Figure 2: **Why "Delayed Gratification" Was an Optical Illusion of the Ruler!** Look at 4 kids: $[\mathbf{2}, \mathbf{4}, \mathbf{1}, \mathbf{3}]$. Before the swap (left), there are **3** out-of-order pairs under **Ruler 1** ($2 > 1$, $4 > 1$, $4 > 3$, green arches) and **1** neighbor cliff under **Ruler 2** ($4 > 1$, red arrow). When 4 and 1 swap to make $[\mathbf{2}, \mathbf{1}, \mathbf{4}, \mathbf{3}]$ (right), **Ruler 1 goes strictly DOWN from** $3 \to 2$ (closer to sorted!), but **Ruler 2 goes UP from** $1 \to 2$ because swapping $4 \leftrightarrow 1$ split one big cliff ($4 > 1$) into two smaller steps ($2 > 1$ and $4 > 3$)!

Figure 2 shows why that happens:

1. **Zero Bubble kids ever cross the wall:** Because Bubble kids only swap with immediate neighbors, if a frozen kid stands at spot #20, **zero kids from the left of** #20 **ever reach the right of** #20!

2. **Two Different Rulers for Messiness:**
   - **Ruler 1 — Total Out-of-Order Pairs $I(v)$ (Green Arches):** Count *every* pair of kids in the whole line where a bigger number is standing to the left of a smaller number. Every single Bubble swap in history reduces $I(v)$ by -1 (**0** uphill steps out of 192 swaps!).
   - **Ruler 2 — Only Neighbor Steps $E(v)$ (Red Arrows):** Only count how many times a kid is bigger than the kid *immediately* to their right. When 4 and 1 swap in $[2, 4, 1, 3] \to [2, 1, 4, 3]$, Total Pairs $I(v)$ goes **down** $(3 \to 2)$, but Neighbor Steps $E(v)$ goes **up** $(1 \to 2)$ because one big cliff $(4 > 1)$ got split into two small steps $(2 > 1$ and $4 > 3)$.
3. **Erik Hoel's "Causal Emergence 2.0" Test Proves Ruler 1 Is the Real Law:** When we test both rulers using Erik Hoel's 2026 *Causal Emergence 2.0* math (*Patterns* 7:101472), **Ruler 1 ($I(v)$) is 100% deterministic with positive Causal Emergence** $(+0.119)$, whereas **Ruler 2 ($E(v)$) has negative Causal Emergence** (-0.253). The kids never walked backwards — Ruler 2 just had a blind spot!

5 Discovery 3: Why the Computer-Science Critics Were Wrong Twice About the "Secret Cliques"!

Remember how the online blog critics claimed: *"The Bubble and Selection kids only cluster together because of textbook Bubble-sort stability on duplicate numbers!"*? Our experiments proved the critics were wrong **twice**:



Figure 3: **How $C = 2$ Copies per Number Proves the Blog Critics Wrong!** Because Selection kids (`S`, orange) scan their target pointer from the left end of the line $(0, 1, 2 \ldots)$, **75% of Selection kids grab the LEFT spots of each number's group, pushing Bubble kids (`B`, blue) to the RIGHT spots. Top (Case A, $C = 2$ copies per number):** Putting `S` on the left and `B` on the right of each 2-kid pair creates the alternating pattern `S B | S B | S B`, which drives clustering **below random** $(0.477 < 0.50$, **Anti-Clustering!)** — the exact opposite of textbook stability! **Bottom (Case B, $C = 8$–10 copies):** Putting all `S` kids on the left half and all `B` kids on the right half (`SSSS | BBBB`) creates huge cliques $(0.659 > 0.50)$.

1. **Test 1 — Zero Duplicate Numbers $(1 \ldots 80$, All Unique):** When every single kid wears a different number, "duplicate-key stability" is impossible because there are no duplicates! Yet

halfway through sorting, the Bubble and Selection kids **still cluster together** ($0.50 \to 0.624$)! Why? Because **Selection kids move like sports cars and Bubble kids move like pedestrians**: Selection kids teleport small numbers straight to the front of the line, shoving a wave of Bubble kids into the right half of the line.

2. **Test 2 — The $C = 2$ "Anti-Clustering" Smoking Gun on Duplicate Numbers (Figure 3):** Even when there *are* duplicate numbers, if textbook Bubble-sort stability were the cause, keeping randomly shuffled equal numbers in their starting order would leave clustering at **0.50** (random) for every group size C. Instead, because Selection kids always scan from the left $(0, 1, 2 \ldots)$, 75% **of Selection kids grab the left half of every number's group** (`S...S | B...B`):

   - When there are **C = 2 copies of each number** (Figure 3, top), putting `S` on the left and `B` on the right of each pair creates `SB | SB | SB`, driving clustering **below** 0.50 (0.477 — **Anti-Clustering!**)!
   - When there are **C = 8–10 copies of each number** (Figure 3, bottom), each group forms `SSSSS | BBBBB`, driving clustering **way up to** 0.659!

6 The Take-Home Lesson: Why Both Biology and Math Win

1. **What Michael Levin Got Right ("Mathematical Free Lunches"):** You really *do* get "more out than you put in" when cells follow simple local rules! Without any central gym teacher or friend-detector code, the geometry of permutations and differences in movement speed automatically create self-sorting, error tolerance, and group segregation. Real biological embryos exploit these mathematical "free lunches" every day.

2. **Why You Don't Need Ghosts in the Machine:** Once you use the right mathematical ruler — **Total Out-of-Order Pairs** $I(v)$ (which has +0.119 Causal Emergence) instead of **Neighbor Steps** $E(v)$ (-0.253), and **sports-car vs. pedestrian traffic waves** (`S...S | B...B`) — 100% of the behavior is explained and predicted by the 6 lines of code!